

02

Tools: OSM and QGIS

UNIVERSITY OF WARWICK

Group formation

Form groups to create the cartographic style (10’)

1. Read all the design briefs
2. Stay around the brief you would like to implement today.
3. Wait for others to join your table/area.
4. When 3 or more people are gathered around, you have formed a group!
    - Introduce yourselves to the colleagues
5. Assign roles:
    - QGIS
    - Documenter
    - Optional:
        - OSM Wiki / taginfo
        - Designer (e.g. icons, colours…)



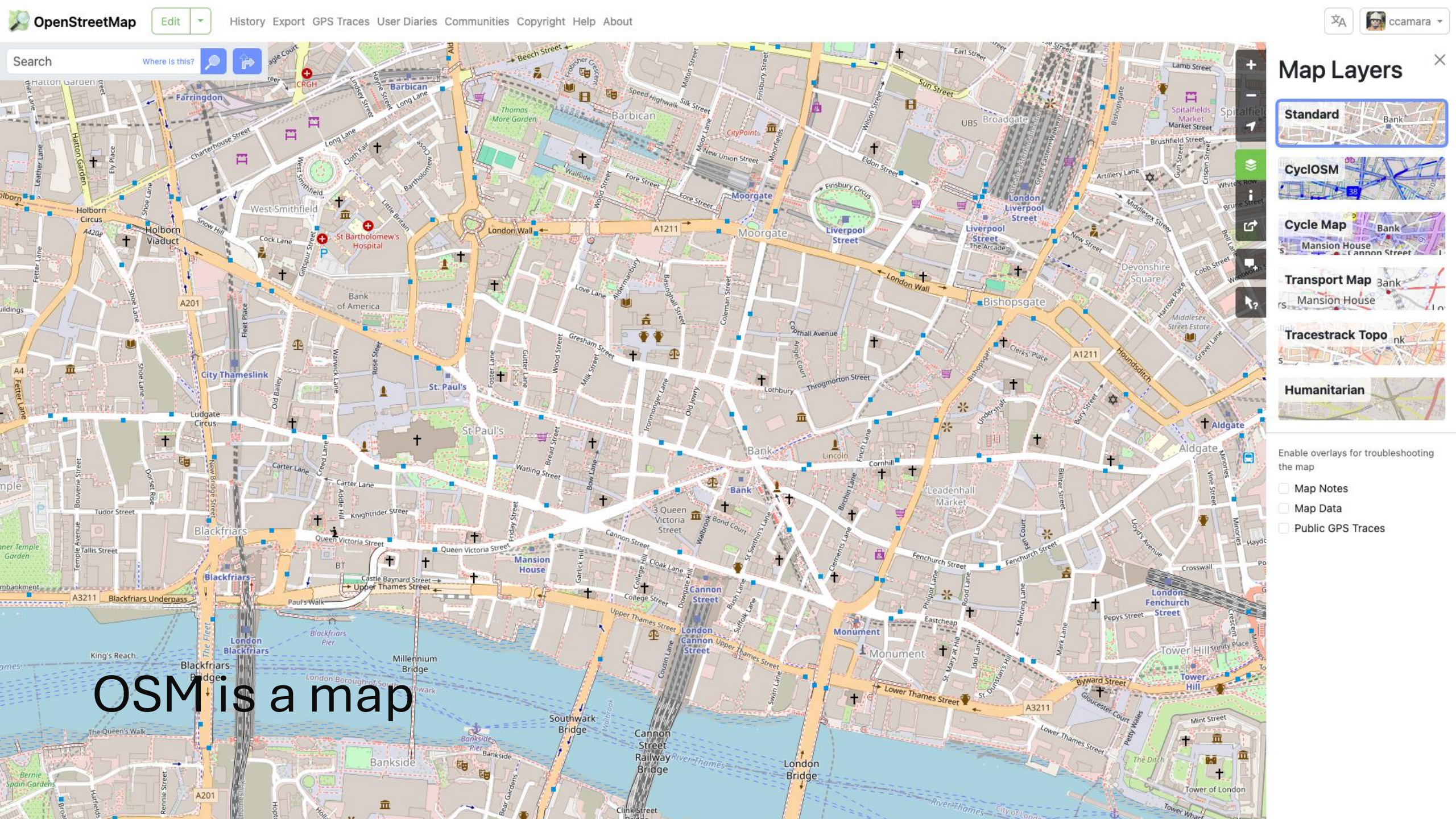
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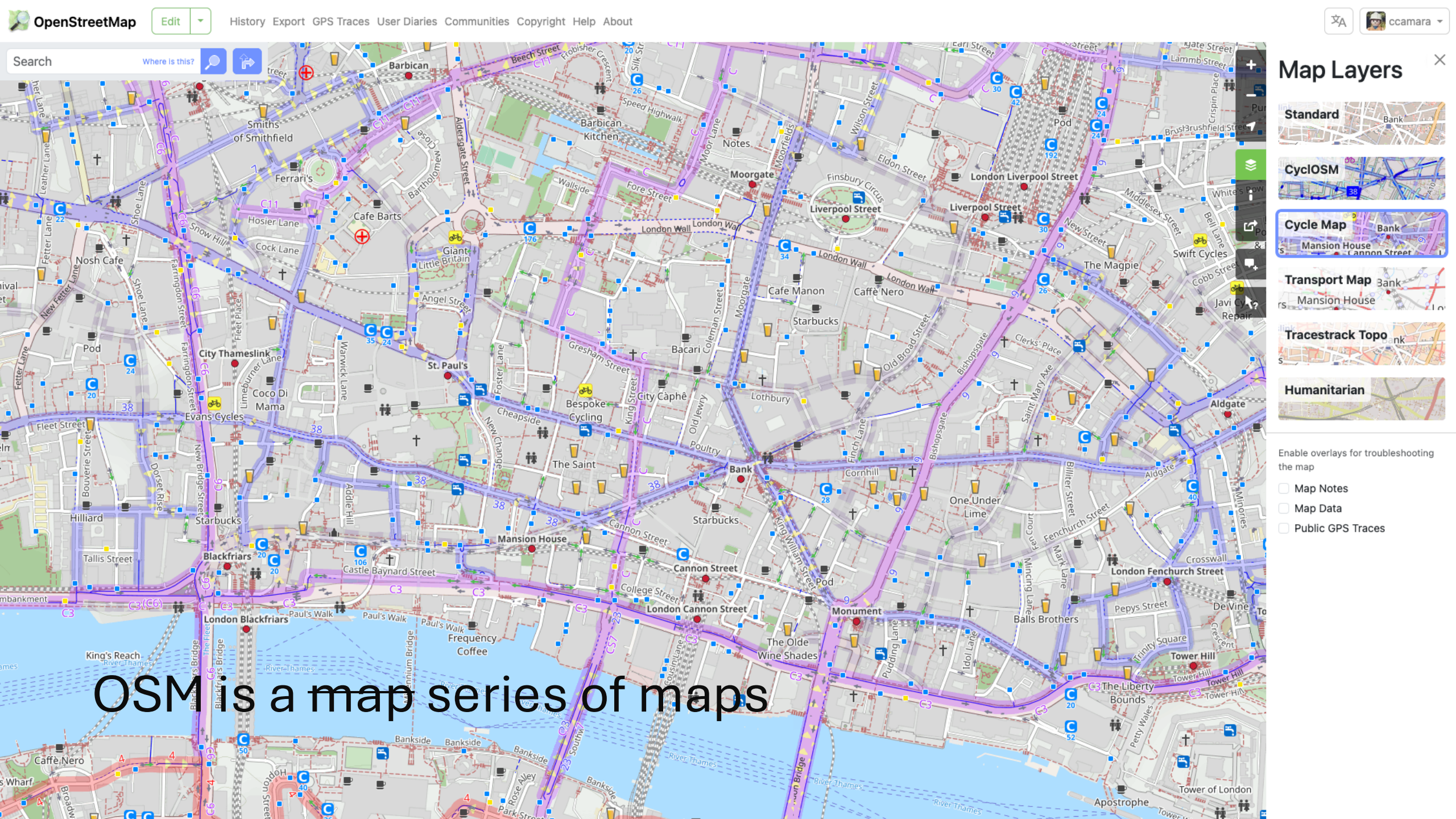


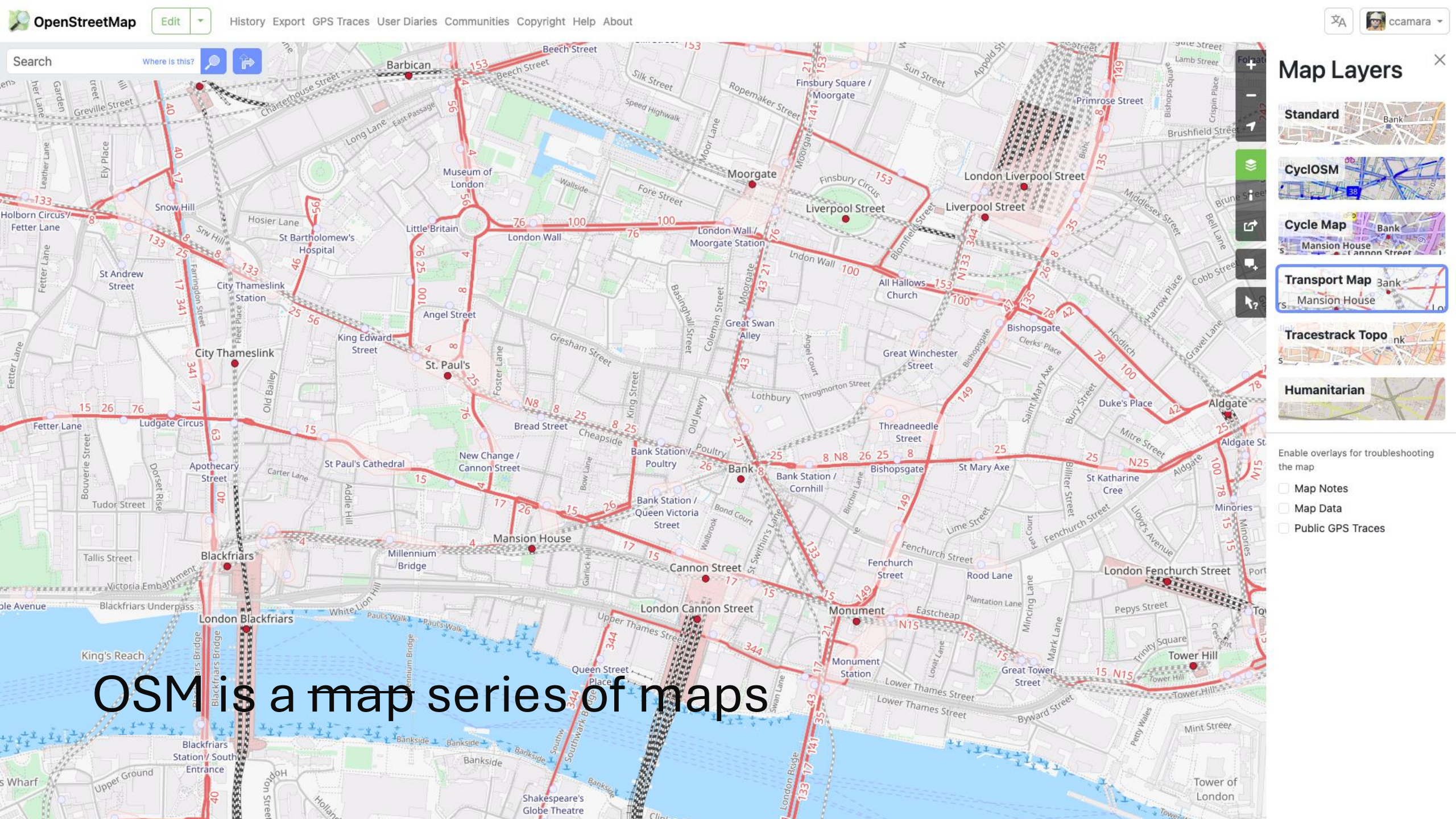
“The Wikipedia of the maps”

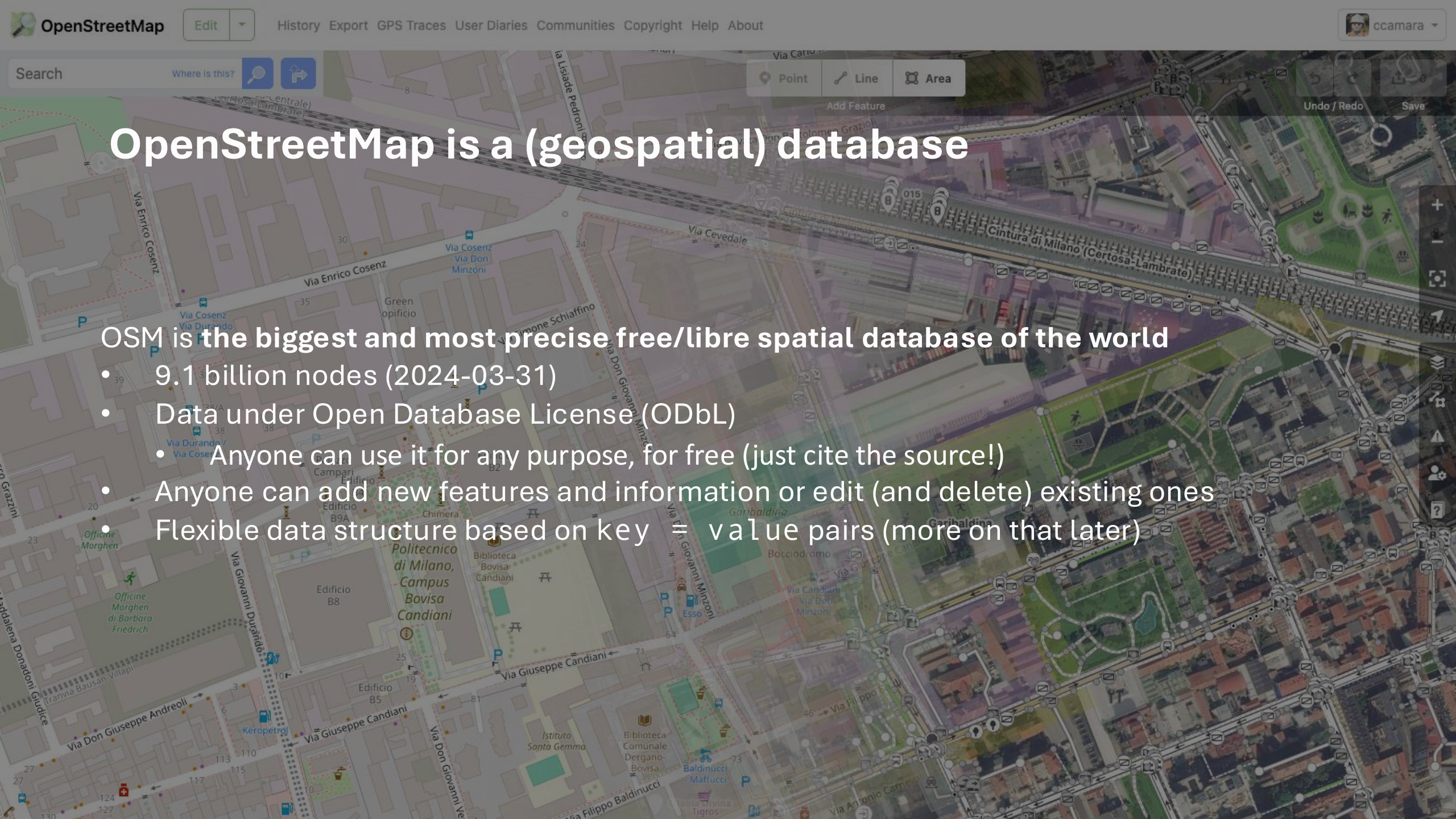
OpenStreetMap

https://openstreetmap.org


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OSM is a ~~map~~ series of maps

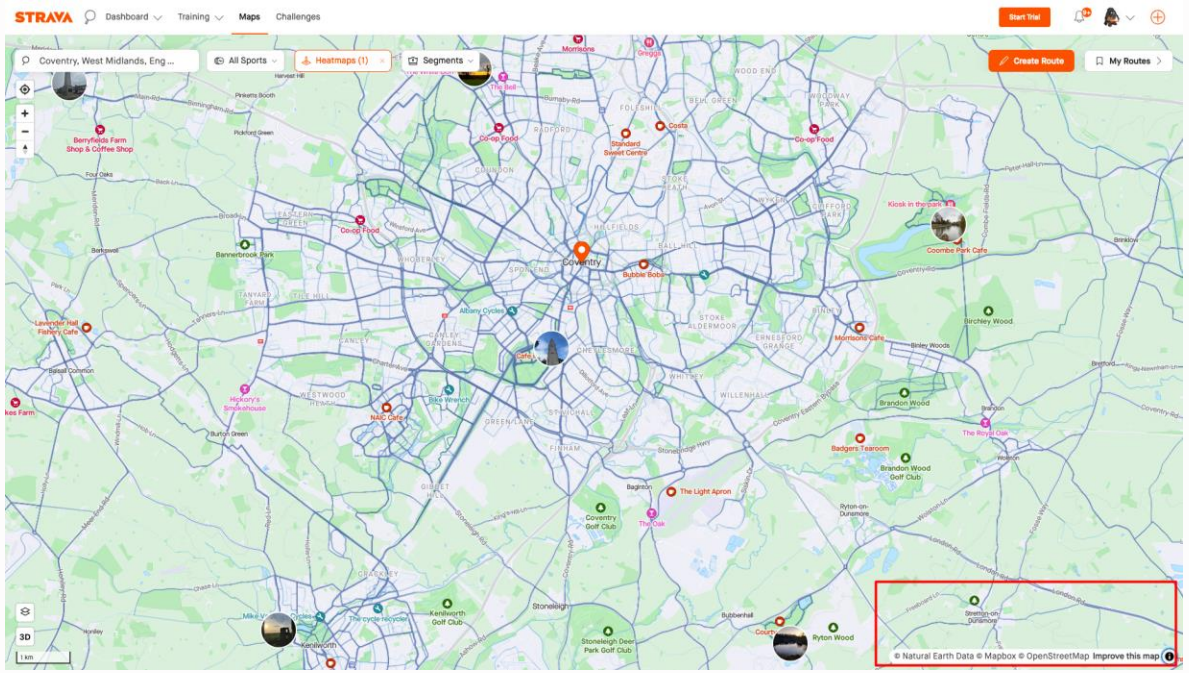

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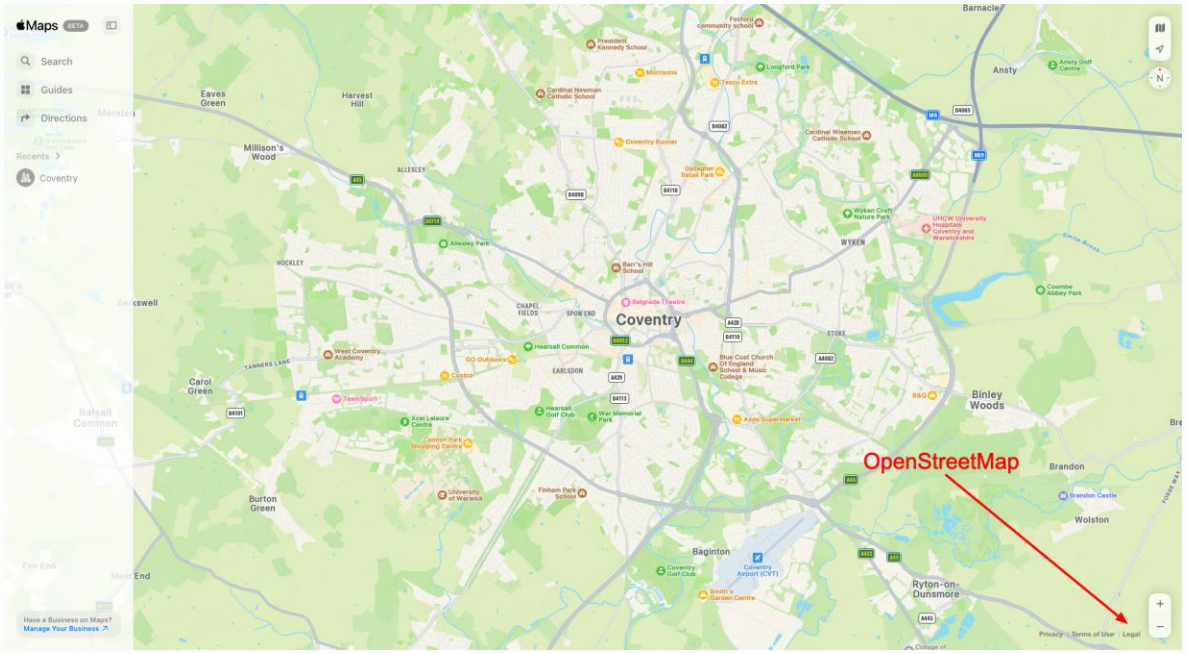


OpenStreetMap is a (geospatial) database

OSM is **the biggest and most precise free/libre spatial database of the world**

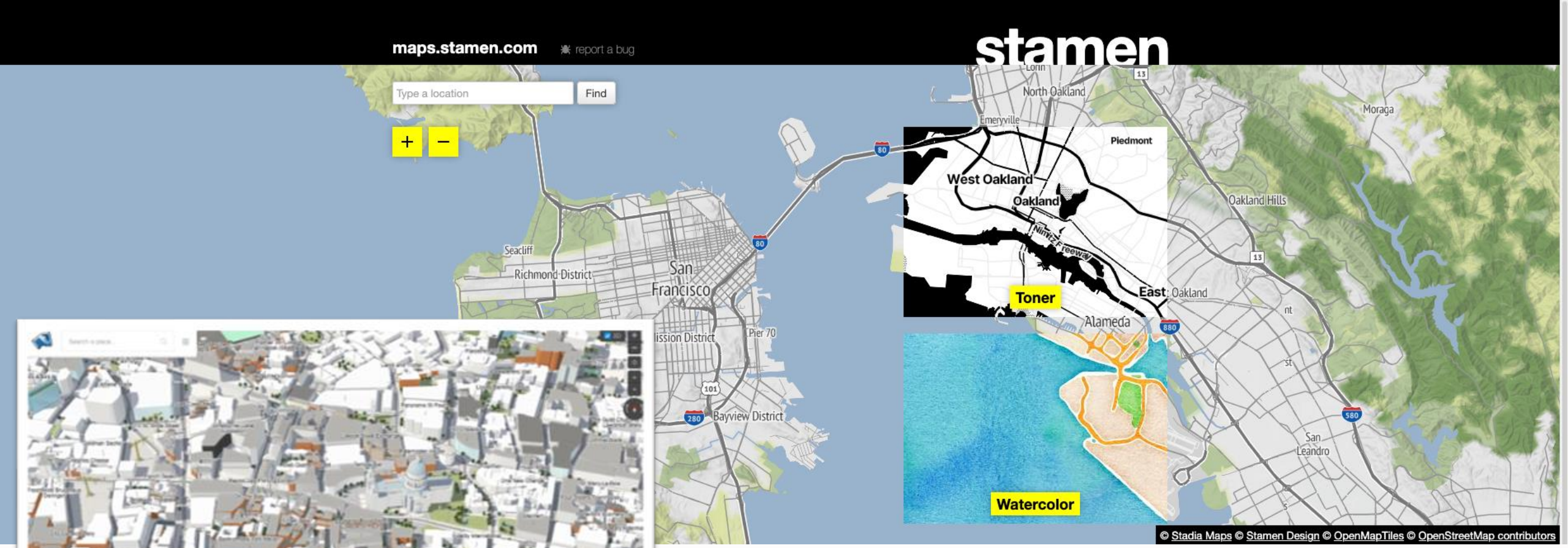
- 9.1 billion nodes (2024-03-31)
- Data under Open Database License (ODbL)
  - Anyone can use it for any purpose, for free (just cite the source!)
- Anyone can add new features and information or edit (and delete) existing ones
- Flexible data structure based on `key = value` pairs (more on that later)





Its data, contributed by 10 million of volunteers, complements official data sources, and **populates thousands of tools and services**.

Major Sites: Amazon, Apple, Baidu Maps, Facebook, Microsoft, Wikipedia and Wikimedia
Transport: Air France, Alaska Airlines, Deutsche Bahn, Grab, SNCF (French rail agency), Uber
Geodata Software and Services: CARTO, Digital Globe, ESRI, Garmin, Mapbox, Telenav
Government: Agence Française de Développement , Government of Brazil, Government of Italy: President's Office, Police Scotland, US National Park Service, US State Department, USAID, Peace Corps....



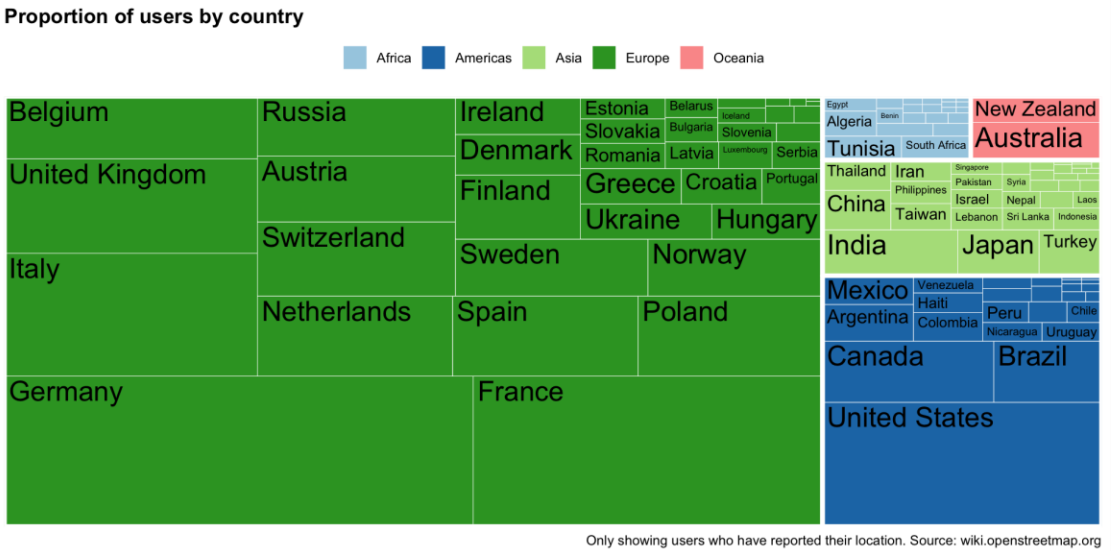


Some of these companies create cartographic styles with OSM data for others to use: Stamen, Carto, Mapbox, Stadia, Mapzen…

OSM is a community (of communities)

More than **10 million users worldwide**, mainly volunteers (but not always)

- All users have **equal level of permissions** to:
  - Add new features to the map
  - Edit existing features to add more detail or remove outdated details
  - Remove
  - Decide what or how to organise the information in the dataset (i.e. what is included and what is not)
- Users may be part of local communities
- The OSM Foundation oversees the OSM project and has several WG



Like Wikipedia, **OSM is based on** principles of **openness** and **neutrality.**

“**OpenStreetMap maps world as it exists**, and includes mapping borders and countries according to actual current situation and **not a preferred or ideal situation**” (OSM Wiki)

What can be mapped?



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Scope

OpenStreetMap welcomes diverse contributions from our diverse community. Defining what falls within OSM remit is, therefore, challenging. As a rule of thumb, anything that ticks all the following conditions would be welcome:

- It is verifiable,
- It is mappable, within the planet Earth,
- It is not private or copyrighted information,
- It is compatible with the OpenStreetMap License,
- It is added manually (see the community guidelines for automated edits),
- It is not listed under Limitations.

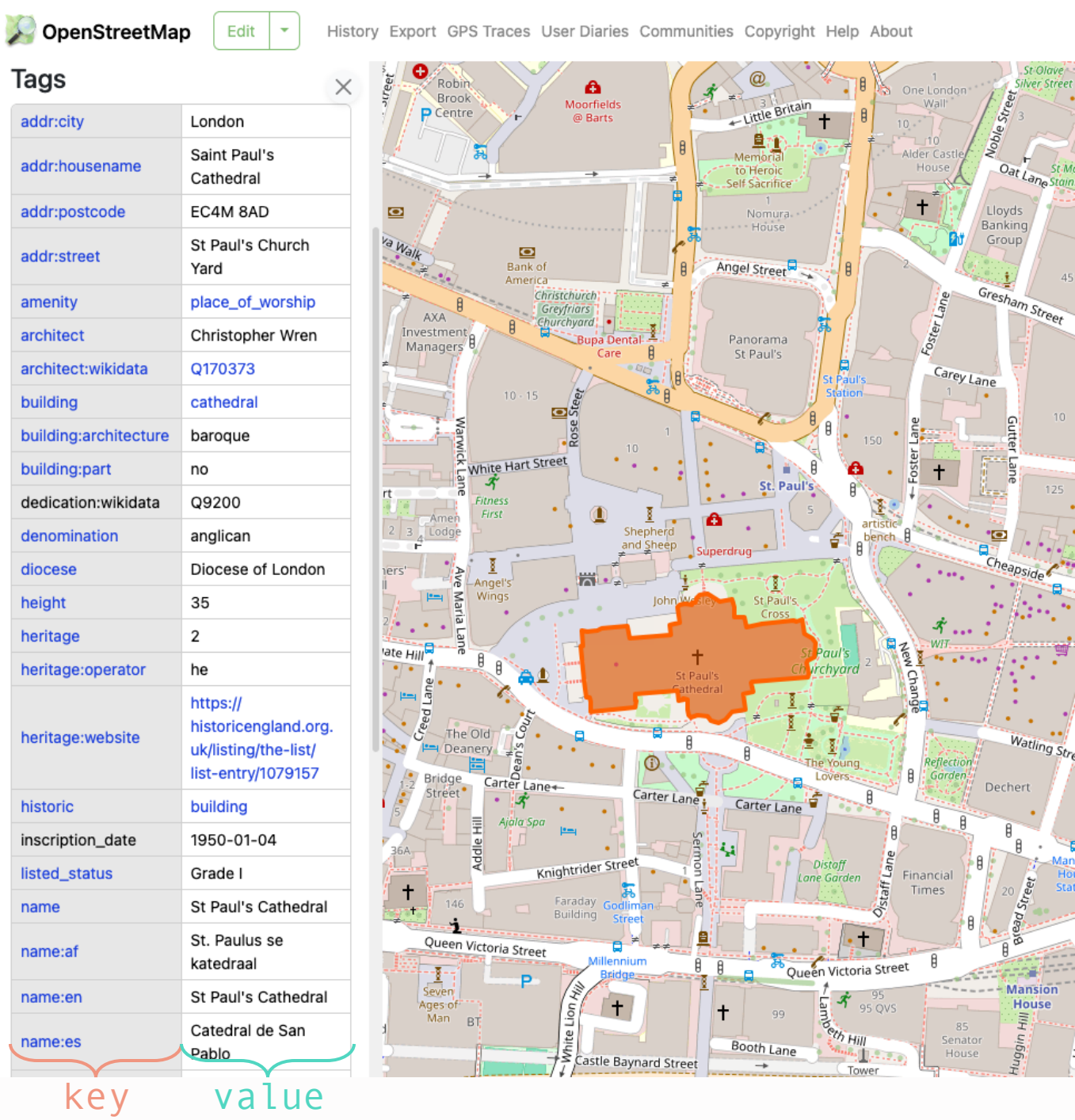
Could these types of data be included in OSM?

- Population’s data (e.g. census)
- Historical data
- Descriptions about a shop
  - menus,
  - Stairs
  - LGTBIQ+ friendliness
  - ratings
- Data from governments, city councils, police…

How are things mapped?

Understanding OSMs data structure

- In OSM, a **feature** is a physical element in the world subject to be mapped (street, building, sign, tree, bench…).
- Each OSM feature is described in two dimensions:
  - Spatial: through their coordinates (latitude and longitude) and element type (point, line, polygon, relationships)
  - **Meaning:** through tags (free format text fields)
- Each **tag** is defined by a pair of `key = value`
  - Keys can be assimilated to categories or topics
  - Value provide detail for the key-specified feature
- Not all keys are displayed on a map (rendered)
  - There's way more information that remains hidden (usually, for good reasons)
  - This information can be queried!



Querying OSM data

Using Overpass Turbo: https://overpass-turbo.eu/

Run | Share | Export | Wizard | Style | Save | Load | Settings | Help | overpass turbo | Map | Data

```
/*
This has been generated by the overpass-turbo wizard.
The original search was:
""Drinking Water""
*/
[out:json][timeout:25];
// gather results
node["amenity"="drinking_water"]({{bbox}});
// print results
out geom;
```



This data can be exported (downloaded) and used to create our own spatial dataset (foundation for a map)

We will be querying Overpass API using a QGIS

How do we know what to query in Overpass Turbo?

In short: OSM Wiki and Taginfo

OSM wiki: collaborative documentation

https://wiki.openstreetmap.org

The wiki describes the consensus reached by OSM community.

Keys and Tags are described in OSM wiki. The wiki explains how to use them:

- Description/meaning
- Applicable elements
- List of accepted values
- Whether it is rendered or not
- ...

We can use the search feature to find how to tag certain features.

The wiki has links to two other important services:

- Overpass Turbo
- Taginfo



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Data item

Tag:amenity=drinking_water

(Redirected from Drinking Water)

Tag:amenity=drinking water · **In other languages** Help

čeština Deutsch **English** español français Nederlands polski português Ελληνικά русский українська 한국어 中文（繁體） 日本語

Other languages...

Drinking water is a place where humans can obtain potable water for consumption. Typically, the water is used for drinking. This includes **drinking fountains** or **bubblers**, water taps, water wells, springs and water points. Generally, this feature provides small quantities of water.

Where drinking water is provided in large quantities, use `amenity=water_point` or `waterway=water_point`; i.e., for refilling tanks in motorhomes, boats, etc.

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1 How to map
  1.1 Useful combinations
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5 External links
6 Common/possible tagging mistakes
7 See also

amenity = drinking_water v·d·e



Description

A drinking water source which provides potable water for consumption.

Rendering in OSM Carto

Group: amenities

Used on these elements

Implies

- drinking_water=yes

How to map [edit | edit source]

Add `amenity=drinking_water` to an existing element or create a new one.

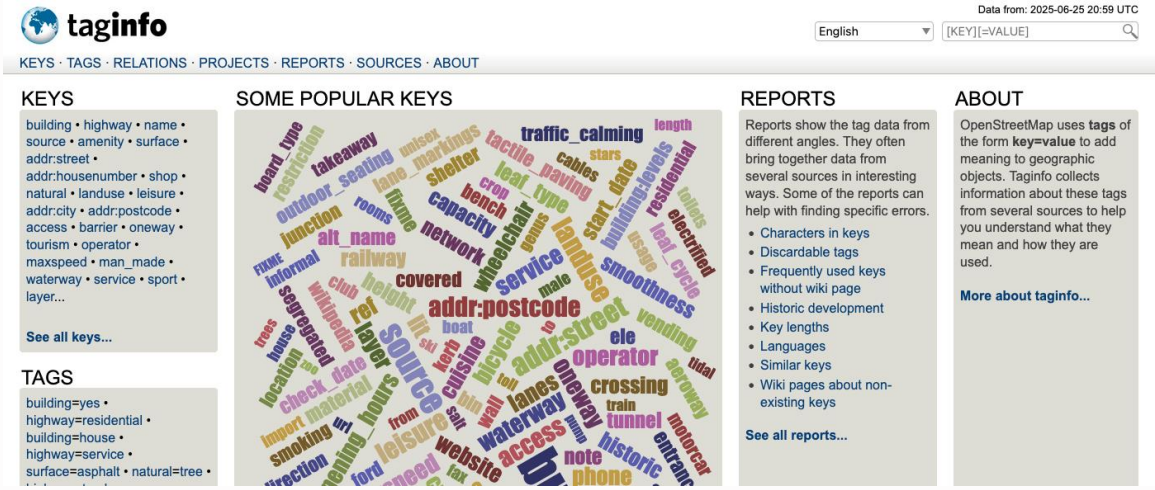
Taginfo

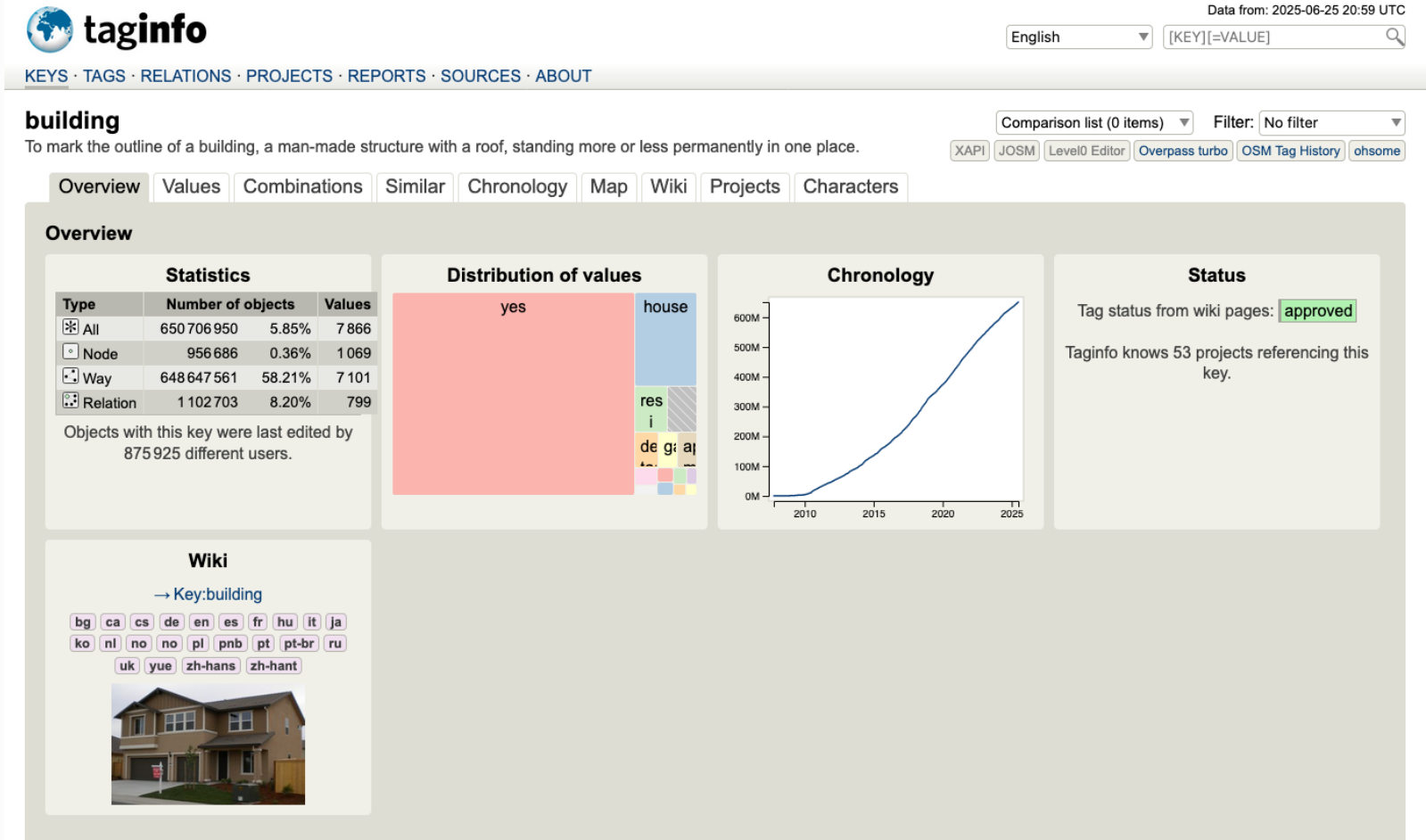
https://taginfo.openstreetmap.org/

Taginfo describes the actual uses of tags in the OSM (de facto).

- Most popular keys
- All possible values for a given key
- When possible, links to the wiki for more information

Linked from OSM wiki pages.







https://qgis.org